TD 3: One-Period Model Step-by-Step Corrections

Stochastic Finance & Risk Modelling

March 18, 2026

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1 Exercise 1: Arbitrage and equivalent measure

The problem asks us to prove that in a one-period model, the existence of an arbitrage opportunity is independent of the choice of equivalent probability measures. Specifically, if P and Q are equivalent probability measures ($P \sim Q$), the market is arbitrage-free under P if and only if it is arbitrage-free under Q.

Let us break down the proof step-by-step.

Step 1: Define an arbitrage strategy under measure Q

Suppose there exists an arbitrage strategy under the measure Q. Let this strategy be denoted by $\xi = (\xi^0, \xi^1, \ldots, \xi^d) \in \mathbb{R}^{d+1}$, where ξ^0 represents the position in the riskless asset, and $\xi^1, \ldots, \xi^d$ represent the positions in the d risky assets.

Let V_t^ξ denote the value of this portfolio at time $t \in \{0, 1\}$. By the definition of an arbitrage strategy under Q, the following three conditions must hold simultaneously:

1. The initial cost of the portfolio is zero:
$$V_0^\xi = 0 \tag{1}$$

2. The portfolio has no risk of losing money (it is almost surely non-negative under Q):
$$Q(V_1^\xi \geq 0) = 1 \tag{2}$$

3. There is a strictly positive probability of making a profit under Q:
$$Q(V_1^\xi > 0) > 0 \tag{3}$$

Step 2: Apply the definition of equivalent measures

We are given that P and Q are equivalent measures. Two probability measures are equivalent if they agree on which events have a probability of zero. Mathematically, for any event $A \in \mathcal{F}$:

$$P(A) = 0 \iff Q(A) = 0 \tag{4}$$

This also directly implies they agree on events that occur almost surely (probability equal to 1). If $Q(A) = 1$, then $Q(A^c) = 0$, which means $P(A^c) = 0$, and thus:

$$P(A) = 1 \iff Q(A) = 1 \tag{5}$$

Step 3: Translate the arbitrage conditions from Q to P

Now, we apply the equivalence property to the conditions from Step 1.

Condition 1 ($V_0^\xi = 0$) is an initial deterministic condition at $t = 0$ and does not depend on the probability measure. It remains exactly the same.

Condition 2 states that $Q(V_1^\xi \geq 0) = 1$. Because P and Q agree on events of probability 1, this immediately implies:

$$P(V_1^\xi \geq 0) = 1 \tag{6}$$

Condition 3 states that $Q(V_1^\xi > 0) > 0$. We know that $P(A) = 0 \iff Q(A) = 0$. By taking the contrapositive, this means $P(A) > 0 \iff Q(A) > 0$. Applying this to the event $\{V_1^\xi > 0\}$, we get:

$$P(V_1^\xi > 0) > 0 \tag{7}$$

Step 4: Conclusion

We have shown that if ξ is an arbitrage strategy under Q, it satisfies:

- $V_0^\xi = 0$
- $P(V_1^\xi \geq 0) = 1$
- $P(V_1^\xi > 0) > 0$

These are the exact definitions of an arbitrage strategy under P. Therefore, if there is an arbitrage under Q, there is also an arbitrage under P.

Because equivalence is a symmetric relationship ($P \sim Q \iff Q \sim P$), the exact same logic can be applied in reverse to show that an arbitrage under P implies an arbitrage under Q. Thus, the market is arbitrage-free under P if and only if it is arbitrage-free under Q.

2 Exercise 2: A complete market

This exercise focuses on the binomial one-period model, which is the foundational building block for pricing options. We have two states of the world at $t = 1$: a "down" state (ω_1) and an "up" state (ω_2).

Question 1: Arbitrage-free condition

We need to prove that the market is arbitrage-free if and only if the risk-free gross return $1 + r$ lies strictly between the down-factor d and the up-factor u. We can prove this by examining the three possible scenarios for $1 + r$.

Case 1: Suppose $1 + r \leq d$

In this scenario, the risk-free rate is so low that even the worst-case return of the stock (d) is better than or equal to the bank's interest rate. We can construct an arbitrage strategy by borrowing money from the bank to buy the stock. Let our strategy be $(\xi^0, \xi) = (-S_0, 1)$. This means we borrow S_0 from the bank (so our cash position is $-S_0$) and buy 1 share of the stock.

- Initial cost: $V_0 = -S_0 + 1 \cdot S_0 = 0$.
- Value at $t = 1$ in the worst case (down state):

$$V_1(\omega_1) = -S_0(1 + r) + S_0 d = S_0(d - (1 + r)) \geq 0 \tag{8}$$

- Value at $t = 1$ in the best case (up state):

$$V_1(\omega_2) = -S_0(1 + r) + S_0 u = S_0(u - (1 + r)) > 0 \tag{9}$$

Since $V_0 = 0$, $V_1 \geq 0$ always, and $V_1 > 0$ with strictly positive probability, this is an arbitrage. Therefore, for the market to be arbitrage-free, $1 + r$ must be strictly greater than d.

Case 2: Suppose $1 + r \geq u$

Here, the bank offers a guaranteed return that is better than or equal to the best-case scenario of the stock. We can construct an arbitrage by short-selling the stock and putting the money in the bank. Let our strategy be $(\xi^0, \xi) = (S_0, -1)$.

- Initial cost: $V_0 = S_0 - 1 \cdot S_0 = 0$.
- Value at $t = 1$ in the best case (up state):

$$V_1(\omega_2) = S_0(1 + r) - S_0 u = S_0((1 + r) - u) \geq 0 \tag{10}$$

- Value at $t = 1$ in the worst case (down state):

$$V_1(\omega_1) = S_0(1 + r) - S_0 d = S_0((1 + r) - d) > 0 \tag{11}$$

Again, this guarantees profits & avoids losses without any initial investment, which is an arbitrage. Thus, we must have $1 + r < u$.

Case 3: Suppose $1 + r \in (d, u)$

By the First Fundamental Theorem of Asset Pricing, a market is arbitrage-free if and only if there exists an equivalent martingale measure (EMM) denoted by Q. Under Q, the discounted stock price must be a martingale:

$$E^Q\left[\frac{S_1}{1 + r}\right] = S_0 \tag{12}$$

Let $q = Q(\{\omega_2\})$ be the risk-neutral probability of the up state. Then $1 - q = Q(\{\omega_1\})$ is the probability of the down state. Expanding the expectation:

$$\frac{1}{1 + r}\left((1 - q)S_0 d + qS_0 u\right) = S_0 \tag{13}$$

Divide both sides by S_0 and multiply by $1+r$:

$$d - qd + qu = 1 + r \implies q(u-d) = 1 + r - d \implies q = \frac{1+r-d}{u-d} \tag{14}$$

For Q to be a valid equivalent probability measure, we must have $0 < q < 1$. Since $1 + r \in (d, u)$, we know $1 + r - d > 0$ and $u - d > 0$, so $q > 0$. We also know $1 + r < u$, meaning $1 + r - d < u - d$, which gives $q < 1$. Thus, a valid EMM exists, proving the market is arbitrage-free.

Question 2: Completeness of the market

The Second Fundamental Theorem of Asset Pricing states that an arbitrage-free market is complete if and only if the equivalent martingale measure Q is *unique*.

From Question 1, we established that avoiding arbitrage requires $1 + r \in (d, u)$. We then derived the risk-neutral probability for the up state:

$$q = Q(\{\omega_2\}) = \frac{1+r-d}{u-d} \tag{15}$$

Because r, d, and u are fixed parameters of the market, there is exactly one mathematical solution for q. Since Q is unique, every derivative can be perfectly replicated by a portfolio of the underlying stock and the riskless bond. Therefore, the market is complete.

Question 3: Call option price

In a complete, arbitrage-free market, the unique price of any derivative π_0 is the discounted expected payoff under the risk-neutral measure Q:

$$\pi_0 = E^Q\left[\frac{(S_1 - K)_+}{1+r}\right] \tag{16}$$

Where $(x)_+ = \max(x, 0)$. We substitute our unique probabilities q and $1 - q$ into the expectation:

$$\pi_0 = \frac{1}{1+r}\left(q \cdot (S_1(\omega_2) - K)_+ + (1-q) \cdot (S_1(\omega_1) - K)_+\right) \tag{17}$$

Substitute $q = \frac{1+r-d}{u-d}$ and $1 - q = \frac{u-1-r}{u-d}$:

$$\pi_0 = \frac{1+r-d}{u-d}\frac{(S_0 u - K)_+}{1+r} + \frac{u-1-r}{u-d}\frac{(S_0 d - K)_+}{1+r} \tag{18}$$

This is the general pricing formula for a call option in a one-period binomial model.

Question 4: Numerical application

We are given: $r = 0$, $S_0 = 100$, $u = 1.2$, $d = 0.9$, $K = 100$.

1. Stock Returns ($\rho(S)$): The return is calculated as $\frac{S_1 - S_0}{S_0}$.

- Up state: $S_1 = 100 \times 1.2 = 120$. Return $= \frac{120-100}{100} = 20\%$.
- Down state: $S_1 = 100 \times 0.9 = 90$. Return $= \frac{90-100}{100} = -10\%$.

So, $\rho(S) \in \{-10\%, 20\%\}$.

2. Call Option Price (π_0^C): First, find the risk-neutral probability q:

$$q = \frac{1 + 0 - 0.9}{1.2 - 0.9} = \frac{0.1}{0.3} = \frac{1}{3} \tag{19}$$

Now, calculate the option payoffs:

- Up state payoff: $(120 - 100)_+ = 20$

- Down state payoff: $(90 - 100)_+ = 0$

The price is the expected discounted payoff ($r = 0$, so discount factor is 1):

$$\pi_0^C = \frac{1}{3}(20) + \frac{2}{3}(0) = \frac{20}{3} \approx 6.67 \tag{20}$$

3. Call Option Returns ($\rho(C)$): The return of the option is $\frac{\text{Payoff}-\text{Price}}{\text{Price}}$.

- Up state return: $\frac{20-20/3}{20/3} = \frac{60/3-20/3}{20/3} = \frac{40/3}{20/3} = 2 = 200\%$.
- Down state return: $\frac{0-20/3}{20/3} = -1 = -100\%$.

So, $\rho(C) \in \{-100\%, 200\%\}$.

Conclusion: The initial investment required for the option (20/3) is much cheaper than the stock (100). However, the option has a massive leverage effect: it offers the potential for much higher percentage gains (200% vs 20%) but also total loss of the invested capital (-100% vs -10%).

3 Exercise 3: Bounds of call option

This exercise establishes the universal no-arbitrage bounds for a call option and then uses a specific model (Poisson distribution) to show that if a market is incomplete, the arbitrage-free price can theoretically be anywhere within those bounds.

1. Universal bounds of a call option

Let π_0 be the arbitrage-free price of the call option today ($t = 0$). By the Fundamental Theorem of Asset Pricing, there exists an equivalent martingale measure (EMM) Q such that the price is the discounted expected payoff:

$$\pi_0 = E^Q\left[\frac{(S_1 - K)_+}{1+r}\right] \tag{21}$$

Upper Bound: We know that for any positive numbers, $(S_1 - K)_+ \leq S_1$ (the option payoff can never exceed the stock price itself). Taking the discounted expectation under Q on both sides:

$$\pi_0 = E^Q\left[\frac{(S_1 - K)_+}{1+r}\right] \leq E^Q\left[\frac{S_1}{1+r}\right] \tag{22}$$

Because Q is a martingale measure, $E^Q[\frac{S_1}{1+r}] = S_0$. Therefore:

$$\pi_0 \leq S_0 \tag{23}$$

Lower Bound: The function $f(x) = (x - \frac{K}{1+r})_+$ is a convex function. We can apply Jensen's inequality, which states that for a convex function f, $E[f(X)] \geq f(E[X])$.

$$\pi_0 = E^Q\left[\left(\frac{S_1 - K}{1+r}\right)_+\right] \tag{24}$$

$$\geq \left(E^Q\left[\frac{S_1}{1+r}\right] - \frac{K}{1+r}\right)_+ \tag{25}$$

Since $E^Q[\frac{S_1}{1+r}] = S_0$, we get the lower bound:

$$\pi_0 \geq \left(S_0 - \frac{K}{1+r}\right)_+ \tag{26}$$

Combining these gives $(S_0 - \frac{K}{1+r})_+ \leq \pi_0 \leq S_0$.

2. Drawing the bounds

If we plot π_0 (y-axis) against S_0 (x-axis):

- The **upper bound** is a straight line starting at the origin with a slope of 1 ($y = S_0$).
- The **lower bound** is 0 for $S_0 < \frac{K}{1+r}$, and then becomes a straight line of slope 1 starting from $S_0 = \frac{K}{1+r}$.
- The valid arbitrage-free prices must fall in the shaded region between these two lines.

3. Poisson model and incomplete markets

We now assume $r = 0$, $S_0 = 1$, and $S_1 \sim \text{Poisson}(1)$ under the historical measure P. Recall that for $S_1 \sim \text{Poisson}(\lambda)$ with $\lambda = 1$:

$$P(S_1 = k) = \frac{e^{-1}1^k}{k!} = \frac{1}{e \cdot k!} \quad \text{for } k \in \{0, 1, 2, \dots\} \tag{27}$$

(a) Show that P is a martingale measure. We need to check if $E^P[\frac{S_1}{1+r}] = S_0$. Since $r = 0$, $E^P[\frac{S_1}{1+0}] = E^P[S_1]$. The expected value of a Poisson(1) random variable is $\lambda = 1$. Because $S_0 = 1$, we have $E^P[S_1] = S_0$, meaning P itself is already a martingale measure!

(b) Show that X and Y_n are Radon-Nikodym derivatives. To define a new valid probability measure Q from P using a random variable Z (where $dQ = ZdP$), Z must be strictly positive and $E^P[Z] = 1$. Let's check $X = e1_{\{S_1=1\}}$:

$$E^P[X] = e \cdot P(S_1 = 1) = e\left(\frac{1}{e \cdot 1!}\right) = 1 \tag{28}$$

Let's check $Y_n = (e - \frac{e}{n})1_{\{S_1=0\}} + (n-1)!e1_{\{S_1=n\}}$:

$$\begin{aligned} E^P[Y_n] &= \left(e - \frac{e}{n}\right) P(S_1 = 0) + (n-1)!eP(S_1 = n) && (29)\\ &= \left(e - \frac{e}{n}\right) \frac{1}{e \cdot 0!} + (n-1)!e\frac{1}{e \cdot n!} && (30)\\ &= \left(1 - \frac{1}{n}\right) \cdot 1 + \frac{(n-1)!}{n!} && (31)\\ &= 1 - \frac{1}{n} + \frac{1}{n} = 1 && (32) \end{aligned}$$

Both have an expectation of 1.

(c) Show that Q_X and Q_n are martingale measures. For Q_X, we must show $E^{Q_X}[S_1] = S_0 = 1$:

$$E^{Q_X}[S_1] = E^P[XS_1] = E^P[e1_{\{S_1=1\}}S_1] = e \cdot 1 \cdot P(S_1 = 1) = e \cdot \frac{1}{e} = 1 \tag{33}$$

For Q_n:

$$\begin{aligned} E^{Q_n}[S_1] &= E^P[Y_nS_1] && (34)\\ &= 0 \cdot P(S_1 = 0) + n \cdot (n-1)!e \cdot P(S_1 = n) && (35)\\ &= n!e\left(\frac{1}{e \cdot n!}\right) = 1 && (36) \end{aligned}$$

Both are martingale measures.

(d) Are they equivalent to P? No. For equivalence, X and Y_n must be strictly positive almost surely. However, $X = 0$ whenever $S_1 \neq 1$. So, events like $\{S_1 = 2\}$ have a positive probability under P but zero probability under Q_X. Therefore, they are not equivalent.

(e) Construction of equivalent martingale measures. Let Q be an EMM (we know at least P is one, from part a). We define $Q_X^\epsilon = \epsilon Q + (1-\epsilon)Q_X$. Because Q is equivalent to P, $Q(A) > 0$ for all non-empty events. For any $\epsilon > 0$, the convex combination $\epsilon Q + (1-\epsilon)Q_X$ guarantees that every event possible under P receives strictly positive probability. The expectation is linear, so it remains a martingale measure.

(f) Is the market complete? No. We just constructed infinitely many equivalent martingale measures by varying ϵ and n. A market is only complete if the EMM is unique.

(g) Option prices at the limits. Let's calculate the option price under Q_X:

$$E^{Q_X}[(S_1 - K)_+] = E^P[X(S_1 - K)_+] = e(1-K)_+P(S_1 = 1) = (1-K)_+ \tag{37}$$

Notice that $(1-K)_+$ is exactly the **lower bound** we found in Question 1 (since $S_0 = 1$ and $r = 0$). As $\epsilon \to 0$, $Q_X^\epsilon \to Q_X$, pushing the option price down to the absolute theoretical minimum.

Now for Q_n:

$$
\begin{aligned}
E^{Q_n}[(S_1 - K)_+] &= E^P[Y_n(S_1 - K)_+] && (38)\\
&= (n-1)!e(n-K)_+ P(S_1 = n) && (39)\\
&= (n-1)!e(n-K)_+ \frac{1}{e \cdot n!} && (40)\\
&= \frac{(n-K)_+}{n} = \left(1 - \frac{K}{n}\right)_+ && (41)
\end{aligned}
$$

As $n \to \infty$, $\frac{K}{n} \to 0$, so the expectation approaches $(1)_+ = 1$. Notice that 1 is exactly S_0, which is the **upper bound** from Question 1. This proves that in this incomplete market, you can find a valid pricing measure that justifies literally any price between the upper and lower no-arbitrage bounds!